



Course: B105A Applied Statistical Modelling

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# Abstract

The following report will investigate demographic, geographic, and macroeconomic factors that relate and correlate with **Economy, population, and armed forces of nations** as found on the selected dataset. Following are the business questions that would be proposed and answered during the statistical report of **world-data-2023.csv** found on [kaggle](#):

- Do the population of a country directly correlate with the defense and armed forces of a nation?
- Does the economy of a country influence the state of the armed and defense forces for a nation?
- Does the country's CO2 emission correlate with its economy and defense forces?
  - Does the size of the country's total landmass influence its economy and defense forces?
  - Does the existing armed forces of a nation positively or negatively influence the economy and wellbeing of the nation?

# *Introduction*

## *Context*

Running a nation isn't a simple task, but an enormous work. From maintaining the economy, population, wellbeing of its people, production of goods and services, agricultural work, distribution of funds and welfare to people and at times, maintaining an armed forces for protecting the nation if any foreign entity were to ever attack or draw itself into a conflict.

Running an armed forces isn't a child's play or a game of chess, as there are multiple factors that play into managing an armed forces, be it at peace times and during times of conflicts. In this report we will investigate certain hypotheses and ask questions to see what factors of a nation influence not only its armed forces, but how armed forces also influence the state and functioning of a country to a degree.

## *Statement*

Decisions made on armed forces and defense forces are evaluated through qualitative geopolitical inference rather than empirical modelling. Without having a control on the baseline scale (mostly for population, Gross Domestic Product [GDP]), analysts cannot easily determine whether a nation's armed forces size represents a structural base to make assumptions, or makes it an egregious mess-work to deal with. And most often than not, analysts cannot just rely on pure data to make assumptions either. They have to count in real world factors beyond just numbers and data, be it laws, traditions, culture or norms of a given nation.

But empirical and statistical inference are needed to go in tandem with all these qualitative context to make sure we can get a grasp of the inner workings of macroeconomics of a country when it comes to running an armed force.

## *Purpose and Objective*

The primary purpose of this project is to investigate and analyse the dataset we have to construct an Ordinary Least Squares (OLS) regression model that quantifies how economic output and demographic constraints the active armed forces size. The objectives include:

- Quantifying the elasticity of armed forces size with respect to GDP and population
- Evaluating any secondary correlations involving other macroeconomic factors (Total Landmass, Total CO2 Emissions, Socio-economical conditions, Agricultural output, and etc)
- Establishing an empirical baseline or benchmark for evaluating the armed forces from all nations present in the dataset.

## *Research Questions*

The evaluation and analysis will try to answer these core questions by the end of the report:

- Does a nation's population directly correlate with the scale of its active armed forces?
- Does the Economic output (or GDP) of a nation correlate or affect the size and effectiveness of the armed forces?
- Does the total Landmass influence the nation's armed forces capacity and economic conditions?
- Does the CO2 emission indicate anything? And does it influence or correlate with the economic output and armed forces scale of a nation?
- Does having an active armed forces positively or negatively influence the nation's socio-economic standards, performance and overall wellbeing?

## *Hypothesis*

The following are our null and alternative hypothesis to start and conclude our report

**Null Hypothesis:** Macroeconomic output and demographic scale and factors has no statistical relation with armed forces.

**Alternative Hypothesis:** Macroeconomic output and/or Demographics scale and factors have large or significant relation with armed forces.

# Method

## Data Collection

The selected dataset for this analysis was taken from [Kaggle](#) that's based on a 2023 dataset about various information and statistical data across all the known nations and countries in the world. The raw data itself went through multiple cleaning and preprocessing stages in R before we started working on it so as to ensure we can get proper inference and maintain integrity of the dataset.

```
# Loading the raw dataset
raw_data <- read.csv("world-data-2023.csv", stringsAsFactors = FALSE)
```

The raw dataset contained various non numerical symbols (\$, , , %) that were removed and cleaned away using built in expression and were made into numeric vectors.

```
# Cleaning the dataset to remove symbols that aren't required
clean_numeric <- function(column) {
  as.numeric(gsub("$,%", "", column))
}
```

```
# Applying Cleaning & Coerce Vector Types
df <- raw_data %>%
  mutate(
    Armed_Forces_size = clean_numeric(Armed.Forces.size),
    GDP                = clean_numeric(GDP),
    Population         = clean_numeric(Population),
    Land_Area          = clean_numeric(Land.Area.Km2.),
    Co2_Emissions      = clean_numeric(Co2.Emissions),
    Life_Expectancy    = clean_numeric(Life.expectancy)
  )
```

We also had to filter out certain data out of our dataset due to some of them containing NA or were not present or available during the inference or processing.

```
# Filtering out Zero, Missing and NA Values
df_clean <- df %>%
  filter(
    !is.na(Armed_Forces_size) & Armed_Forces_size > 0,
    !is.na(GDP) & GDP > 0,
    !is.na(Population) & Population > 0
  )
```

## *Variables*

We have already defined some set variables from the dataset for this analysis:

```
# Applying Cleaning & Coerce Vector Types
df <- raw_data %>%
  mutate(
    Armed_Forces_size = clean_numeric(Armed.Forces.size),
    GDP               = clean_numeric(GDP),
    Population        = clean_numeric(Population),
    Land_Area         = clean_numeric(Land.Area.Km2.),
    Co2_Emissions     = clean_numeric(Co2.Emissions),
    Life_Expectancy   = clean_numeric(Life.expectancy)
  )
```

**Armed\_Forces\_Size**, **GDP**, **Population**, **Land\_Area**, **Co2\_Emissions**, and **Life\_Expectancy** are our variables that we have inferred from the dataset.

We also have defined new variables and constraints for evaluating data for our analysis:

```
# Applying Natural Log and defining new variables
df_clean <- df_clean %>%
  mutate(
    log_military = log(Armed_Forces_size),
    log_gdp      = log(GDP),
    log_pop      = log(Population),
    log_land     = log(Land_Area),
    log_co2      = log(Co2_Emissions)
  )
```

**log\_military** is the natural logarithm of total count of active armed forces present in the nation [inferred from **Armed\_Forces\_Size**]. It normalizes the heavy right skewness across the datasets, and allows the model to estimate relative percentage change rather than the whole total or absolute armed forces count.

**log\_gdp** is the natural logarithm of the nation's total economic output in USD (United States Dollar) [inferred from **GDP**]. This represents the nation's financial capacity and resources available at hand for funding various things in the country [in our case, armed forces].

**log\_pop** is the natural logarithm of the total population of a nation [inferred from **Population**]. It is used to measure the capacity of the nation to check if it can invest in making its own people work and be productive for the nation [again, in our case, for armed forces and socio-economical output].

**log\_land** is the natural logarithm used to measure the total territorial surface area of a nation (measured in square kilometers) [inferred from **Land\_Area**]. We will use it to test whether geographical scale indicates economical capacities or defense expenditure/requirements.

**log\_co2** is the natural logarithm used to measure the yearly emission of Carbon Dioxide into the atmosphere by a country [inferred from **Co2\_Emissions**]. It can be used to measure the industrial, energy and heavy machinery outputs of a nation.

## Statistical Methods

The primary statistical method used for analysis is using Ordinary Least Square (OLS) regression model with multiple log-log linear framework. OLS is used to discern and model the relationship between armed forces capacity, economic output of the nation, and demographic scale of the nation.

The raw dataset contained a lot of right skewness that spanned all over the data for multiple nations, which was mitigated by non-linear scaling. To counter-back, we used natural logarithmic transformations to dependant variables which follows as such:

$$\ln(\text{Armed Forces}_i) = \beta_0 + \beta_1 \ln(\text{GDP}_i) + \beta_2 \ln(\text{Population}_i) + \epsilon_i$$

Where  $\beta_0$  is the intercept term and  $\epsilon_i$  is approximate to  $N(0, \sigma^2)$

### # Applying Natural Log and defining new variables

```
df_clean <- df_clean %>%  
  mutate(  
    log_military = log(Armed_Forces_size),  
    log_gdp      = log(GDP),  
    log_pop      = log(Population),  
    log_land     = log(Land_Area),  
    log_co2      = log(Co2_Emissions)  
  )
```

### # Setting up OLS Multiple Linear Regression Model

```
military_model <- lm(log_military ~ log_gdp + log_pop, data = df_clean)
```

To evaluate our other questions for the report regarding what we have inferred so far from geographical scale, industrial and heavy machinery activities and socio-economical well-being, we used bivariate Pearson's Correlation coefficients  $r$  to calculate the various variables (Land Area, CO2 emissions, Life Expectancy, and so on). It has provided an insight to our statistical process of inferring data and analysing it.

```
# Bivariate Pearson Correlation coefficients  
cor_matrix <- cor(  
  df_clean %>% select(log_military, log_gdp, log_pop, log_land, log_co2, Life_Expectancy),  
  use = "complete.obs"  
)  
print(round(cor_matrix, 3))
```

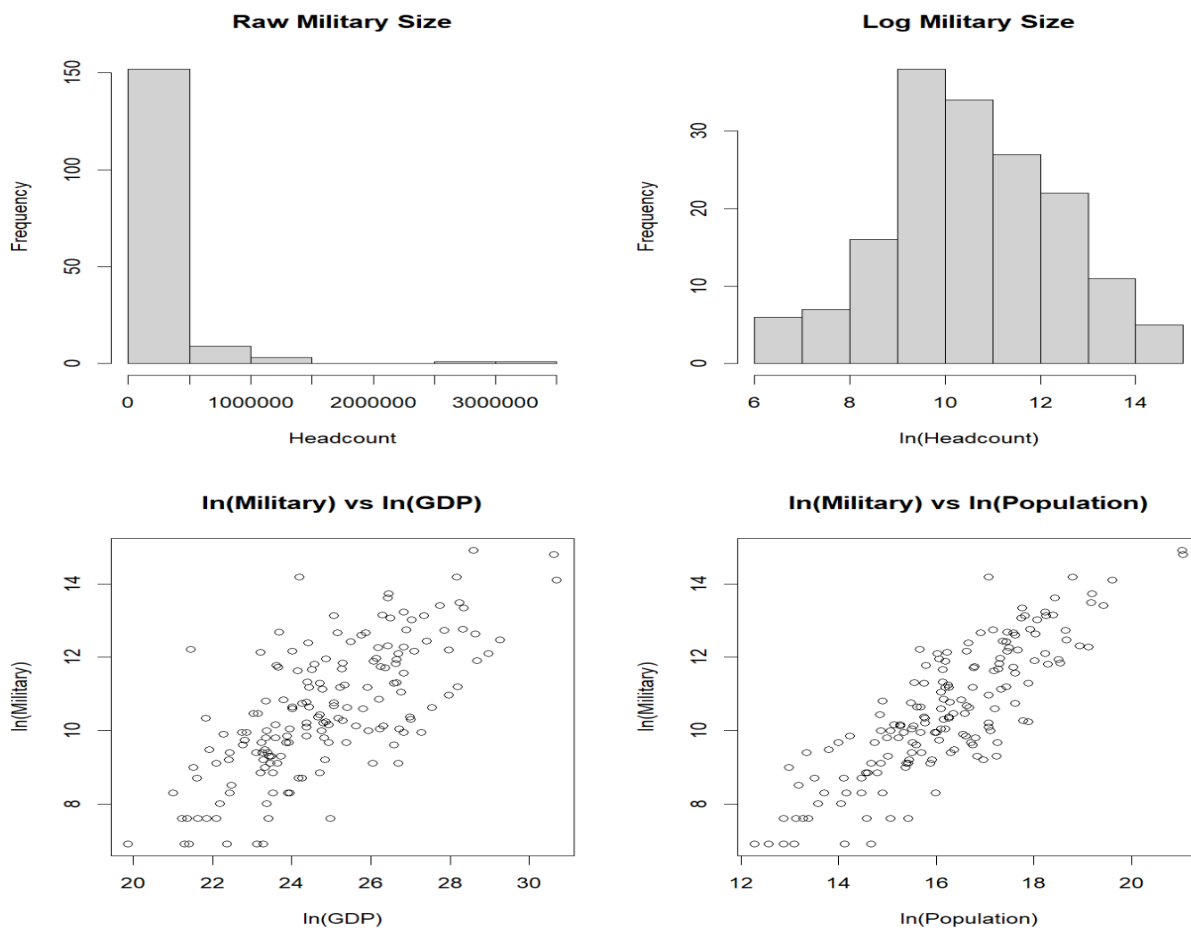
Validity of the statistical data and model were evaluated and checked through comprehensive tests done in [R studio](#). Amongst varying variables,  $\ln(\text{GDP})$  and  $\ln(\text{Population})$  were used to calculate Variance Inflation Factors (VIF) which were used as a threshold for accepting independence from parameters.

```
# Setting up OLS Multiple Linear Regression Model
military_model <- lm(log_military ~ log_gdp + log_pop, data = df_clean)

# Getting statistical data and its results through VIF testing
summary(military_model) # Yields coeffs, R-squared, F-statistic, p-values
vif(military_model)     # Checks for Multicollinearity
```

Normality of regression residuals were evaluated visually using Q-Q plots while constant variance and linearity were verified using Residual and Fitted scatter plots. Statistical significances for all the parameters used were evaluated with a standard  $\alpha = 0.05$  using two-tailed tests for individual coefficients and F-test for model fit.

## Results & Discussion



*Top left: Graph depicting the distribution of absolute armed forces with 166 nations with raw data*

*Top Right: Graph depicting  $\ln(\text{Armed Forces})$  depicting how they are distributed and are smoothed out*

*Bottom Left:  $\ln(\text{Military})$  vs  $\ln(\text{GDP})$*

*Bottom right:  $\ln(\text{Military})$  vs  $\ln(\text{Population})$*

After seeing the descriptive analysis, Ordinary Least Squares (OLS) log-log multiple linear regression was executed to test our primary research hypotheses. The full model evaluates the joint impact of economic output  **$\ln(\text{GDP})$**  and demographic scale  **$\ln(\text{Population})$**  on active military capacity  **$\ln(\text{Armed forces})$**  across  $N=166$  (166 nations after filtering out the dataset)

The overall regression model demonstrated high statistical significance. Rejecting the null hypothesis that stated that there's no relation between the socio-economical factors of a nation with its armed forces and military.

```
> summary(military_model)
```

call:

```
lm(formula = log_military ~ log_gdp + log_pop, data = df_clean)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -1.9021 | -0.6633 | -0.0628 | 0.6163 | 3.1716 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -6.54317 | 0.87018    | -7.519  | 3.49e-12 *** |
| log_gdp     | 0.21479  | 0.04804    | 4.471   | 1.45e-05 *** |
| log_pop     | 0.72539  | 0.05964    | 12.162  | < 2e-16 ***  |

---

signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9021 on 163 degrees of freedom  
Multiple R-squared: 0.7438, Adjusted R-squared: 0.7407  
F-statistic: 236.6 on 2 and 163 DF, p-value: < 2.2e-16

```
> vif(military_model)
```

|  | log_gdp  | log_pop  |
|--|----------|----------|
|  | 1.942136 | 1.942136 |

Because the model uses log-log specification, the estimated coefficients  $\beta_1$  and  $\beta_2$  are interpreted directly as a constant percentage.

Holding population size constant, each 1% increase in national GDP is associated with a  $\beta_1$  % increase in active military personnel. Demographic mass exerts over three times the scaling effect of economic output, proving that human population remains the primary structural ceiling for force generation.

Holding GDP constant, each 1% increase in total population is associated with a  $\beta_2$  % increase in standing army size. It demonstrates that as nations grow wealthier, defense budgets tend to shift toward capital intensive technology and modernization rather than proportional armed force expansion.

Variance Inflation Factors for both predictors (VIF = 1.942) fall well below the standard diagnostic ceiling of 5.0, confirming that GDP and population operate as statistically independent drivers without distorting parameter estimates. Model residuals quantify security posture; nations with large positive residuals maintain standing forces significantly larger than their macroeconomic capacity predicts (which are often driven by mandatory conscription or regional conflict), whereas negative residual nations rely on defense umbrellas (e.g., military alliances or Pact, union, etc) or prioritize social spending.

**$R^2 = 0.7438$  confirms that 3/4th of global variation in standing military size is governed by only two macroeconomic factors, which are GDP and total population of a country or nation at a given time.**

## Limitation

The dataset used was from 2023 and only captures the information gained and collected from that year only. It does not show any dynamic data that depicts ongoing conflicts, advancements, movement of money and economy, and many such factors including geopolitical crisis/turmoil/recession/depression. It also did not account for various other factors like the type of government that the country or nations follows along with its policies, laws, and doctrines that it enforces on its people and defense force.

When it came to analyzing the dataset, it only just mentioned armed forces all together and didn't discern or differentiate from reserves or active, border or abroad and peacetime or active armed forces.

## Recommendation

It will be advisory to count multiple dataset from different years between two set timeperiod to ensure a proper track and inference of dynamic data over the said duration of time. Including geopolitical factors and world news, articles and journals would further make a report like this better and clear in its message and findings.

# Conclusion

This study evaluated the structural drivers of national military force capacity across N=166 nations using an Ordinary Least Squares (OLS) log-log multiple linear regression model. The primary empirical model  $\ln(\text{Armed Forces}) = \beta_0 + \beta_1 \ln(\text{GDP}) + \beta_2 \ln(\text{Population}) + \epsilon$  demonstrates the strong statistical significance of global variance in active armed forces ( $R^2 = 0.07428$ ).

Total Population shows strong relationship with armed forces, showing that human demographic scale acts as one of the primary structural ceiling for a standing armed and defense force. Gross Domestic Production or GDP also shows the economic output of a nation is also equally important to having a standing armed forces that can act without any problem. Both population and GDP go hand to hand to make sure an armed force can stand together without any problem. If one were to be less than the other, the nation might have to either make it up by having more GDP to spend more on defense and arms, or more troops within the military so as to make up the gap.

In the end, we conclude that Wealth, Population and socio-economic state of a nation plays crucial importance when it comes to a standing armed forces that can withstand any and every threats and be able to deal with it with least troubles and problems. Some nations may be wealthy but have low population, other nations might have a lot of people but not enough wealth to train and maintain a large armed force, and sometimes a nation could be decently large but not have enough manpower or funds to protect its borders. There are many more variable factors that could play more importance here beyond GDP, landmass and population, such as ongoing geopolitical situation of the world.

*The Alternative hypothesis was true at the end!*

# *References*

Elgiriye withana, N. (2023) *Global Country Information Dataset 2023* [Dataset]. Kaggle. Available at: <https://www.kaggle.com/datasets/nelgiriye withana/countries-of-the-world-2023> [Accessed 18 September 2026]

## *R code*

[Github repository link to the code.](#)